Programs Calories Members

ACEest FITNESS & GYM

Programs Calories Workouts Members Gym Info Versions

ACEest FUNCTIONAL FITNESS SYSTEM

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DevOps Assignment 1 — Flask Web API with Full ACEest Business Logic

Evolved from 10 desktop versions → modern CI/CD pipeline

🏋️ Training Programs

Explore structured workout and nutrition plans tailored for your fitness goals.

🔥 Fat Loss

Burn fat with high-intensity workouts and calorie deficit diet.

💪 Muscle Gain

Build strength and size with progressive overload training.

🌱 Beginner

Start your fitness journey with balanced workouts and simple plans.

🔥 Calorie Calculation Logic

Daily calorie targets are calculated using a simple formula:

calories = weight (kg) × factor

FL: 22 MG: 35 BG: 26

Calculate Daily Calorie Target

Body Weight (kg)

e.g. 75

Program

Select Program

Calculate

📋 Workout Log

Logged Workouts 0

No workouts logged yet.

Log a Workout

Workout Name

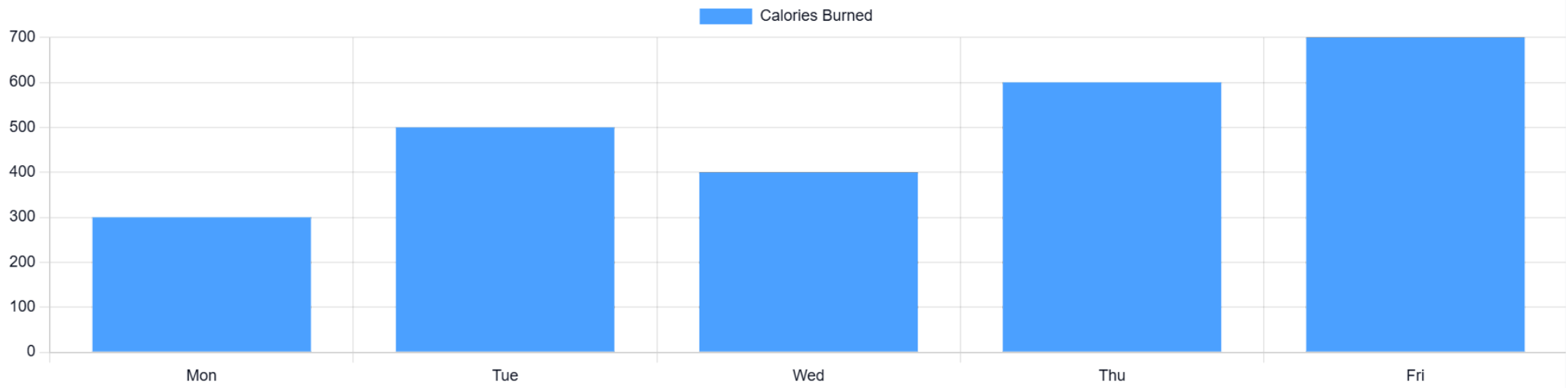
e.g. Back Squat 5×5

Duration (minutes)

e.g. 60

Log Workout

📊 Workout Analytics



Members

Registered Members 0

No members registered yet.

Register Member

Full Name

e.g. Arjun Kumar

Email

e.g. arjun@aceest.in

Program (optional)

— select program —

Register Member

🏢 Gym Capacity & Metrics

150	10,000	250
Max Capacity	Gym Area	Break-even
users	sq ft	members

🔌 API Endpoints

Method	Endpoint	Description
GET	`/`	Browser UI (this page)
GET	`/health`	Health check — returns `{"status":"healthy"}`
GET	`/programs`	All 3 ACEest programs with workout & diet plans
GET	`/programs/FL`	Single program by code (FL / MG / BG)
POST	`/calories`	Calculate daily kcal: `{"weight_kg":75,"program_code":"FL"}`
GET	`/workouts`	List all logged workouts
POST	`/workouts`	Log a workout: `{"name":"...","duration_minutes":60}`
GET	`/members`	List all registered members
POST	`/members`	Register a member: `{"name":"...","email":"..."}`
GET	`/gym-info`	Gym capacity, area, break-even metrics

📜 Version History (10 Desktop → 1 Web API)

The application evolved from Tkinter desktop scripts to this Flask web service deployed via Docker and CI/CD.

- **Aceestver-1.0** — Tkinter GUI, 3 programs (FL/MG/BG), workout & diet display, gym metrics
- **Aceestver-3.0.1** — Tabbed UI (Client Summary + Progress & Analytics), AI program generator, PDF reports

- **Aceestver-1.1** — Added calorie factor formula, colour-coded programs, CSV export
- **Aceestver-2.1.2** — SQLite database for client profiles and weekly progress tracking
- **Aceestver-2.2.1** — Matplotlib charts for adherence visualisation
- **Aceestver-2.2.4** — Height/weight schema, membership expiry, full DB migration logic
- **Aceestver-3.1.2** — Role-based login screen (Admin / Trainer), full workout & exercise logging
- **Aceestver-3.2.4** — Multi-table DB (users, clients, workouts, exercises, metrics), membership billing
- **Aceestver1.1.2** — Multi-client table UI, progress chart, notes field, CSV bulk export
- **Aceestver2.0.1** — Clean DB-backed final desktop baseline before web migration
- **app.py (this)** — Flask REST API + Bootstrap UI + Docker + GitHub Actions + Jenkins ✅

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